Routed Experts Are Not Redundant: Training Dynamics and Branch Proportion in Small Mixture-of-Experts Models

AUTHOR NAMES
AFFILIATION
email@domain

***Abstract*—Small Mixture-of-Experts language models built with an always-on shared expert appear to under-use their routed experts: removing the entire routed branch costs little in perplexity, while removing the single shared expert costs a great deal. We show that this asymmetry reflects optimization dynamics and architectural proportion rather than any property of the computation being performed. Across models of 58M to 436M parameters trained on a domain-labelled corpus, the imbalance is transient: in the 134M reference model the two branches equalize by roughly 300M tokens, while deeper models remain shared-dominated longer. We trace the imbalance to a gradient-exposure race in which the shared expert, updated on every token, outcompetes routed experts updated on a k/E fraction of them. The race is causally controllable: slowing the shared expert reallocates computation to the routed branch at no cost in perplexity, and the reallocation persists to the end of our budget even when the handicap covers only the first tenth of training. Where the branches settle is governed jointly by their relative width and by the routing granularity, not by the number of experts. At fixed routing, an eightfold change in the shared expert's width moves the shared branch's share of ablation impact from 0.28 to 0.81 at essentially unchanged perplexity; at identical widths, top-1 and top-4 routing land at 0.97 and 0.17; and single-expert routing never engages the routed branch at any width we tested.**

***Index Terms*—mixture of experts, training dynamics, sparse models, language models, expert specialization**

I. Introduction

Mixture-of-Experts (MoE) layers replace a dense feed-forward block with a router that dispatches each token to a small number of experts. The DeepSeekMoE design [1] adds an always-on *shared* expert beside the *routed* ones, on the intuition that the shared expert absorbs computation common to all tokens and thereby frees the routed experts to specialize. Because small MoEs are inexpensive to train and every token can be labelled, they have become a common testbed for studying how experts divide work.

In this small-scale regime a striking observation recurs. Ablating the entire routed branch barely moves held-out perplexity, whereas ablating the lone shared expert degrades it severely. Read at face value, the measurement says the routed experts are nearly useless, and it invites two explanations: either small MoEs are over-parametrized on the routed side, or the shared expert actively suppresses the routed ones. Both explanations attribute the imbalance to the architecture.

We argue that neither is right, and that the imbalance says almost nothing about what the computation requires. It is instead the joint product of two things that have no obvious connection to the task: how quickly each branch is trained, and how wide each branch is. The apparent redundancy is transient, disappearing once the model has seen a few hundred million tokens. Its cause is a race for gradient signal that the shared expert wins by construction, and that a single learning-rate multiplier hands to the routed branch instead, at no cost in perplexity. Applying the multiplier over the first tenth of training suffices; once the race is decided, the allocation persists long after the handicap is removed. Where the two branches finally come to rest is set jointly by their relative width and by the routing granularity, and not, as the mechanism naively suggests, by the number of routed experts: holding the routed branch identical down to its parameter shapes and varying only the shared expert's width moves the shared branch's share of ablation impact from 0.28 to 0.81, while at identical widths top-1 and top-4 routing land at 0.97 and 0.17. Granularity is the one lever among these that has to be paid for, and single-expert routing marks its degenerate end, never engaging the routed branch at all.

Taken together these results reframe the shared/routed split as a largely free design choice. Among configurations of comparable size, held-out perplexity varies by less than 0.1 while the branch split spans 0.25 to 0.81, so a practitioner may place the computation more or less where they wish. That freedom matters whenever the routed experts are themselves the object of interest, as they are in interpretability work, expert pruning, and modular editing [2], [3], and it means that a diagnosis of expert redundancy drawn from an undertrained checkpoint is not merely imprecise but misleading. Whether experts so engaged also become better objects of study turns out to depend on which lever is used: widening the routed branch relative to the shared one more than doubles their domain specialization, whereas the learning-rate control does not reliably move it at all.

II. Related Work

Conditional computation through a sparsely gated MoE layer was introduced by Shazeer et al. [4] and scaled by GShard [5] and Switch Transformer [6], which popularized the auxiliary load-balancing loss in the form we use; ST-MoE [7] added the router z-loss for stability. Our reference configura-

tion is top-2, in the style of Mixtral [8], and Section VI-A varies that choice.

The always-on shared expert originates with DeepSeek-MoE [1] and carries through DeepSeek-V2 and V3 [9], [10] and further open families such as Qwen2 [11], motivated by the argument that isolating common knowledge in a shared branch lets the routed experts specialize more sharply. DeepSeekMoE pairs it with a second argument, that finer expert segmentation sharpens specialization; Section VI-A tests a controlled analogue of that manipulation in isolation. Both arguments concern what the branches represent. Our results concern how the branches are trained and proportioned, and they suggest that at small scale the latter consideration dominates.

Observational studies of expert behaviour report that experts organize more by token identity and surface form than by topic, both in Mixtral [8] and across training in OLMoE [2]. These studies characterize the outcome of training. We add a controlled causal probe and identify a training-dynamics cause for the small-scale imbalance that observational analysis alone cannot isolate.

The mechanism we identify instantiates a broader phenomenon: the early phase of training commits a network to structure that later training refines rather than revisits [12], [13]. Routing instability in that same phase motivated StableMoE [14], and recent theory studies when a mixture's experts diverge [15]. Our hysteresis experiment gives the commitment claim an unusually direct test, a single knob released mid-training, at the level of whole branches.

III. Experimental Setup

a) Model: The model is a decoder-only MoE implemented from scratch in PyTorch, with Multi-head Latent Attention [9], RoPE [16] and RMSNorm [17]. Each layer carries E routed experts and one shared expert with top-k routing. The reference configuration has $E = 8$, $k = 2$, 8 layers, hidden size 512, context 512 and 134M parameters; every expert, routed or shared, has intermediate width 1024. Section VI varies the two widths independently and writes them I per routed expert and I_{shared} for the shared one. Configurations range from 58M to 436M parameters. All runs use AdamW (β_1=0.9, β_2=0.95, weight decay 0.1) with cosine decay to zero after 1500 warmup steps, batch 64 at sequence length 512, peak learning rate 3×10^{-4}, an auxiliary load-balancing coefficient of 10^{-2} and a router z-loss of 10^{-3}, in bfloat16, with a capacity factor of 1.25 during training and no capacity dropping during analysis; 30 000 steps correspond to 983M tokens. Every experiment fits on one RTX A6000.

b) Data: Pre-training streams three token-labelled corpora: FineWeb-Edu [18] (60%), Cosmopedia-v2 [19] (25%) and OpenWebMath [20] (15%), tokenized with a 32k vocabulary and packed to 512 tokens. Held-out evaluation uses a skip-offset of the same streams rather than a fully disjoint split.

c) Measuring where the computation lives: We ablate a whole branch, either all routed experts or the shared expert, and record the resulting increase in held-out perplexity. Ablating a single expert is uninformative under top-k routing because the router simply reallocates its tokens to the survivors; only whole-branch ablation isolates a branch's contribution. We summarize the two measurements by the shared branch's share of total ablation impact,

$$s = \frac{\Delta_{\text{shared}}}{\Delta_{\text{shared}} + \Delta_{\text{routed}}}, \tag{1}$$

so that $s = 0.5$ denotes branches of equal importance and $s \to 1$ a shared-dominated model. Almost every trajectory we measured dips to a minimum and then rises slowly, the exceptions being the top-1 configurations of Section VI-A, which decline monotonically and never leave shared dominance; we therefore report the minimum reached over the trajectory and, where the endpoint matters, the value at 983M tokens. Every configuration for which we ran a second seed or a replicate run (eight, across both depths) put the spread on s at 0.03 or less, an order of magnitude below the effects discussed below.

TABLE I
Branch ablation impact over training, four seeds; mean ± std on the ablation columns.

tokens	PPL	remove shared	remove routed	s
98M	4.18	$+3.44 \pm 0.17$	$+0.82 \pm 0.03$	0.81
197M	3.63	$+2.37 \pm 0.17$	$+1.58 \pm 0.06$	0.60
295M	3.41	$+1.99 \pm 0.14$	$+1.98 \pm 0.06$	0.50
491M	3.20	$+1.99 \pm 0.16$	$+2.25 \pm 0.01$	0.47
983M	3.00	$+2.45 \pm 0.18$	$+2.48 \pm 0.11$	0.50

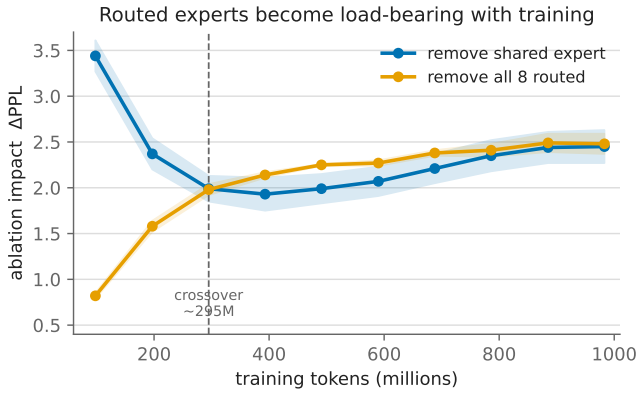


Fig. 1. Ablation impact of removing the shared expert against all eight routed experts, over a four-seed trajectory at constant batch size. Bands are ±1 standard deviation. The branches cross to equal importance by roughly 295M tokens.

IV. Branch Importance Is a Function of Training Tokens

We train the reference model for 983M tokens with four seeds and measure branch importance at every checkpoint. Table I and Fig. 1 show the result.

At 98M tokens the shared expert accounts for 81% of the ablation impact and the routed experts do indeed look close to redundant. This is the regime in which many pilot studies

TABLE II
SHARED-EXPERT LEARNING-RATE MULTIPLIER, TWO SEEDS, MEASURED AT 98M TOKENS.

m	remove shared	remove routed	s	PPL
1.0 (baseline)	$+3.46 \pm 0.12$	$+0.85 \pm 0.01$	0.80	4.16
0.3	$+2.48 \pm 0.02$	$+1.94 \pm 0.00$	0.56	4.10
0.1	$+0.72 \pm 0.08$	$+4.46 \pm 0.11$	0.14	4.10

operate, including our own earlier experiments. By 295M tokens the branches have equalized, and they remain balanced through 983M; routed specialization grows in parallel, with the expert–domain normalized mutual information (NMI, defined in Section VII) rising from 0.0008 at initialization to 0.0174. All four seeds follow the same trajectory with tight error bars.

The practical reading is narrow but firm. Branch importance is not a property of the architecture that can be measured once; it is a function of how much data the model has seen. Conclusions about routed redundancy drawn below a few hundred million tokens do not survive further training, and the effect is large enough that the sign of a comparison can change.

V. A Gradient-Exposure Race

Why should the shared expert lead at all? The asymmetry is present in the update rule before it is present in the weights. Writing the block output on token t as

$$y_t = f_{\text{sh}}(x_t) + \sum_{e \in \text{topk}(t)} g_{t,e}\, f_e(x_t), \tag{2}$$

with f_{sh} the shared expert, f_e the routed ones and $g_{t,e}$ the router's gate weight for expert e on token t, a batch of N tokens delivers gradient from all N of them to the shared parameters but from only about Nk/E to any given routed expert. Under a common learning rate the shared expert therefore accumulates roughly E/k times more effective updates per parameter during the early phase, when gradients are large and the allocation of work is still plastic. The residual stream compounds this: it routes common computation to whichever submodule reduces loss fastest, which further starves the routed experts of useful gradient.

This account makes two predictions. Reducing the shared expert's learning rate should let the routed experts capture computation earlier, and the crossover should scale with E/k. The first holds; the second does not, and Section VI is devoted to what replaces it.

A. A learning-rate handicap decides the race

We place the shared expert's parameters in their own optimizer group at $\text{lr} \times m$ and leave everything else untouched. Table II reports the state at 98M tokens, deep in the regime where the baseline looks shared-dominated.

Slowing the shared expert makes the routed branch load-bearing far earlier: a fivefold increase in routed ablation impact at a fixed token budget, shifting the crossover from roughly 295M tokens to below 200M at $m = 0.3$ and to before training

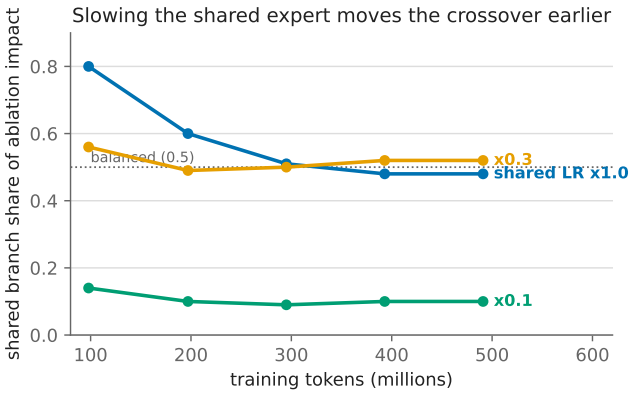


Fig. 2. Shared branch share against training tokens for three shared-expert learning-rate multipliers, two seeds. Slowing the shared expert moves the crossover earlier, and the shift grows with the strength of the handicap.

settles at $m = 0.1$; Fig. 2 traces the full trajectories. The effect is monotone and dose-dependent. It refutes the hypothesis that routed experts simply mature on their own schedule; they were being out-competed.

The multiplier runs train on a 15 000-step schedule whose cosine decay is correspondingly compressed; one tenth of the way in, where Table II is read, the two schedules' global learning rates differ by under three percent, so the contrast is the multiplier. The baseline row is the two-seed reading of the reference run, which is why it differs in the last digit from Table I's four-seed row. Held-out perplexity is unchanged across the three conditions, and if anything marginally better when the shared expert is slowed. The shared expert is therefore not necessary for the computation it performs in the baseline. The model places the work there because that is where it can be learned fastest, not because that is where it belongs.

B. The control holds at greater depth

A mechanism that depended on our particular depth would be of little interest, so we repeat the sweep on a model of twice the depth, 16 layers and 251M parameters. The control reproduces almost exactly. At 98M tokens the shared share falls from 0.82 to 0.64 to 0.21 across the same three multipliers, against 0.80, 0.56 and 0.14 at 8 layers, and perplexity is again unchanged. The right panel of Fig. 3 places the two depths side by side. (The sweep's own baseline reads 0.82 where the 983M-token baseline run reads 0.84 at the same point, a run-to-run difference within the spread reported in Section III.)

What depth does change is the timing of the unaided crossover, and it changes it in the direction that makes the phenomenon more treacherous rather than less. The 8-layer baseline crosses to equal importance at roughly 295M tokens and stays there. The 16-layer baseline never crosses within our budget: it reaches a minimum of 0.54 near 491M tokens and then drifts back up to 0.59 by 983M, as the left panel of Fig. 3 shows. That baseline is a single run, though its

sweep-side replicate above tracks it to within 0.02 over their common range. Deeper models therefore remain in the shared-dominated regime for considerably longer, which is what the mechanism predicts, since more residual capacity gives the shared expert's head start more room to compound. It also means the warning of Section IV grows stronger with scale rather than weaker: the larger the model, the longer an observer will be able to look at it and conclude, wrongly, that its routed experts are redundant. We emphasize that this concerns the crossover reached without intervention; the following subsection shows the deeper model can be driven well past balance and kept there.

C. A temporary handicap is enough

If the split is decided by an early race, a handicap covering only that race should suffice, and the allocation should not revert once it is lifted. We test this on the 16-layer model by ramping m linearly from 0.1 back to 1.0 by a chosen step, after which every parameter trains at the base rate. Fig. 4 shows the outcome for four ramp endpoints.

The reallocation does not undo. Once the handicap is removed the share stays near where the handicap left it rather than returning to the baseline trajectory, and the shared branch reasserts itself only slowly. The response is monotone in the duration of the handicap: the share at 983M tokens is 0.33, 0.26, 0.21 and 0.19 for handicaps lifted at 98M, 197M, 295M and 491M tokens respectively, against 0.59 with no handicap at all and 0.14 for a handicap never lifted. Final perplexities are 2.88, 2.88, 2.88 and 2.89, against 2.90 for the baseline and 2.91 when the handicap is never released.

Three features of that curve matter more than its monotonicity. A brief handicap already does much of the work: lifting it after 98M tokens, one tenth of the budget, reaches 0.33, which is more than half of the distance from the baseline to the never-released limit. The increments then shrink, by 0.07, 0.05 and 0.02 across the successive endpoints, but the never-released limit at 0.14 lies a further 0.05 below the longest ramp, so the response flattens without reaching a plateau anywhere in the range we swept. And releasing the handicap is not merely harmless but marginally preferable: every ramped condition ends at 2.88 or 2.89, better than both the untouched baseline and the permanently handicapped run, which is the only condition here that is worse than doing nothing. The knob is thus a schedule intervention rather than a standing tax on the shared expert, which is what a race-then-commit dynamic predicts, since what wins the early race is what the residual stream subsequently builds upon.

We replicated the 295M endpoint with a second seed and obtained the same summary statistics, a minimum of 0.14 and 0.21 at 983M tokens. The other four conditions are single-seed, and the sweep covers one depth and one handicap strength, so we report the shape of the dependence rather than a functional law.

TABLE III
TWO DISENTANGLING CONTRASTS, BOTH AT E=8, k=2. VALUES ARE THE MINIMUM SHARED SHARE REACHED OVER THE TRAJECTORY.

Varying routed width, $I_{\text{shared}} = 1024$			
I	256	1024	2048
$s_{\min}$	0.79	0.47	0.37
parameters	58.2M	133.7M	234.4M
Varying shared width, $I = 1024$; routed branch identical			
I_{shared}	512	1024	4096
$s_{\min}$	0.28	0.47	0.81
parameters	127.4M	133.7M	171.5M

VI. WHAT SETS WHERE THE SPLIT LANDS

The mechanism explains why the shared expert leads and shows that the lead can be reversed. It does not say where the split comes to rest, and the natural prediction from the gradient-exposure argument, that the resting point should follow E/k, turns out to be wrong. We assemble eleven 8-layer configurations, all at top-2 routing: the five runs of Table III, a capacity-matched sweep over expert count described below ($E \cdot I$ held at 8192 for $E \in \{4, 8, 16, 32\}$), and three runs from an earlier sweep at ratio 1.0 ($E = 4$, 16 and 32 at $I = I_{\text{shared}} = 1024$, 83M to 436M parameters). Ten trained to 983M tokens; the earlier $E = 4$ run stopped at 491M with its share flat from 295M on. Across all eleven, expert count is nearly irrelevant and the ratio of the two branches' widths orders every run. Section VI-A then asks what happens when the routing itself is changed.

Table III isolates the two candidates. Both contrasts hold $E = 8$ and $k = 2$ fixed, so expert count, routing and per-expert gradient share are constant throughout. The second contrast is the decisive one: with I pinned at 1024 the routed branch is identical across the three runs, the same number of experts of the same width receiving the same routed capacity $k \cdot I = 2048$, and yet the split moves by 0.53. Whatever governs the resting point, it is not the absolute capacity of the routed branch.

What the two contrasts share is the ratio I/I_{shared}. Grouping every configuration by that ratio yields bands that do not overlap, as Fig. 5 shows: shared shares of 0.72–0.81 at ratio 0.25, 0.63 at ratio 0.5 (a single configuration), 0.47–0.56 at ratio 1.0 and 0.25–0.37 at ratio 2.0. The ordering survives an eightfold range in ratio, an eightfold range in expert count and a sevenfold range in parameter count.

Expert count does not explain the residual spread within a band, and it does not even act in a consistent direction: at ratio 2.0 more experts raise the share, from 0.25 at $E = 4$ to 0.37 at $E = 8$, while at ratio 0.25 more experts lower it, from 0.79 at $E = 8$ to 0.72 at $E = 32$. A reader might still suspect that the sweep confounds expert count with capacity, since a naive sweep over E at fixed per-expert width grows the routed branch as it grows E. Holding the product $E \cdot I$ at 8192 removes that confound and equalizes parameter counts; perplexity is then flat across $E \in \{4, 8, 16, 32\}$ at 2.99, 3.00, 3.00 and 3.02, while the share continues to track the ratio that the same manipulation necessarily varies. At matched capacity,

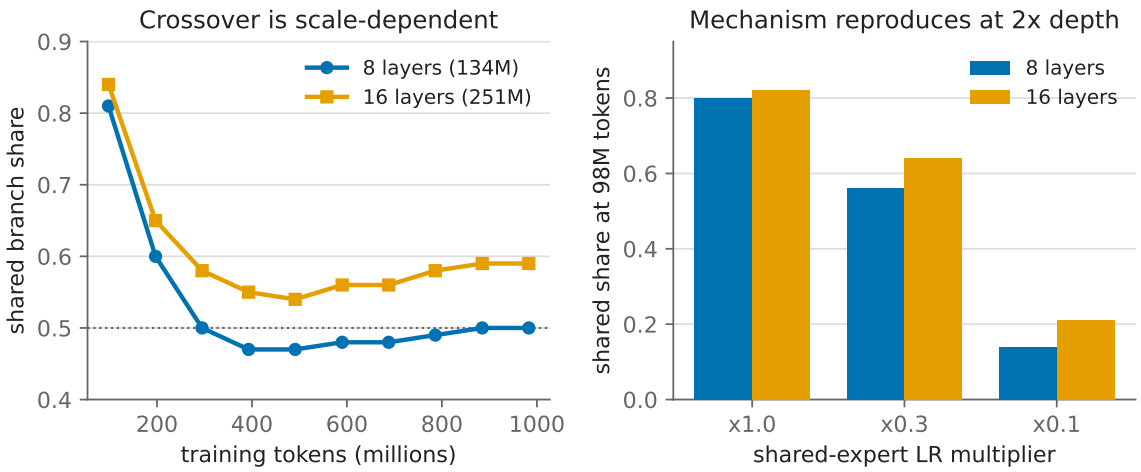


Fig. 3. Left: shared branch share against tokens for the unaided baseline at two depths. The 8-layer model crosses to equal importance; the 16-layer model approaches but does not, within 983M tokens. Right: the learning-rate control reproduces at twice the depth, lowering the share dose-dependently at both scales.

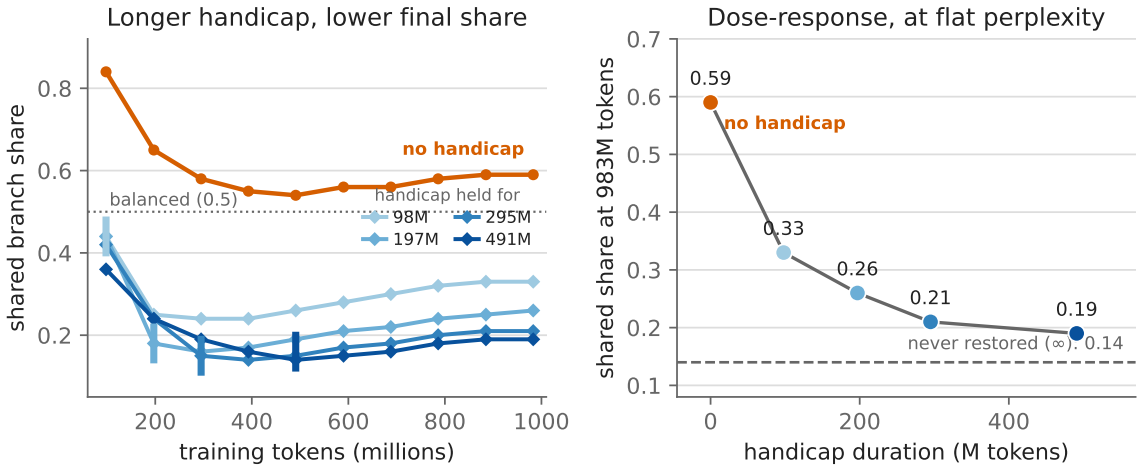


Fig. 4. Left: shared branch share against training tokens at 16 layers, for four handicap durations; the tick marks the point at which each handicap was lifted. Right: the resulting dose-response, final share against how long the handicap was held, with the never-released limit as a dashed reference. Every ramped condition ends within 0.01 perplexity of the others.

expert count is thus close to a free parameter for perplexity, and it moves the split only within the residual spread of a ratio band.

A. *Does routing granularity enter?*

Every configuration above uses top-2 routing, so the ordering just established is, strictly, an ordering at fixed k. Because a token passes through k routed experts but only one shared expert, k has an obvious claim to matter, and we test it directly by holding the widths at $I = I_{\text{shared}} = 1024$, so the width ratio is 1.0 throughout, and varying only k.

Table IV shows that k enters, and that it is the strongest single lever in this study: moving from top-1 to top-4 at identical widths swings the split from 0.97 to 0.17, further than the 0.53 obtained by quadrupling the shared expert's width. The width-ratio result of the preceding subsections must therefore be read as holding at fixed routing, which is how we state it.

TABLE IV
ROUTING GRANULARITY AT FIXED WIDTHS (E=8, I=I_{shared}=1024, RATIO 1.0), TWO SEEDS EACH (k=2: FOUR). THE SPLIT IS THE MINIMUM OVER THE TRAJECTORY; PERPLEXITY AND NMI ARE AT 983M TOKENS.

k	1	2	4
$s_{\min}$	0.97	0.47	0.17
final PPL	3.13	3.00	2.99
routed domain NMI	0.0003	0.0174	0.0235

This also refines, rather than simply refutes, the E/k prediction of Section V. That prediction fails on the E axis, as the

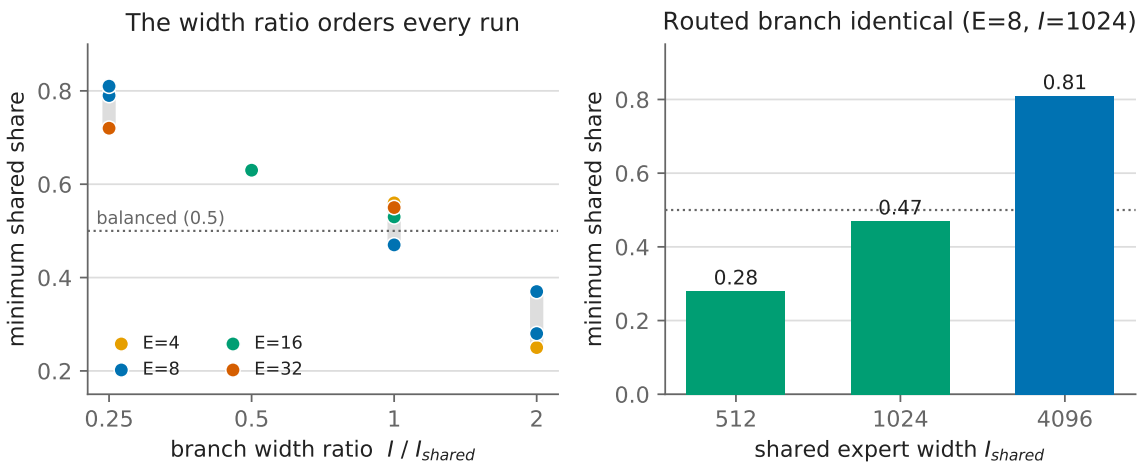


Fig. 5. Left: minimum shared share against the branch width ratio, one point per configuration, coloured by expert count; the bands at each ratio do not overlap. Right: the decisive contrast, in which the routed branch is identical across all three runs and only the shared expert's width changes.

preceding subsection showed: at a fixed width ratio, varying E from 4 to 32 leaves the split within the single band 0.47–0.56 and not even in a consistent direction, so per-expert gradient share cannot be what governs the resting point. It succeeds on the k axis, and in the predicted direction, since raising k lowers the shared share. The asymmetry is informative. What fails is the *ratio* k/E; what survives is k alone. The two runs below show that k's effect is not even carried by the extra width it activates, which leaves the granularity of the routing itself, how many experts share a token, as an operative variable alongside the branch proportion.

Unlike the width ratio, this lever is not free. Top-1 costs 0.13 in perplexity and top-4 doubles the routed floating-point cost per token, whereas the width manipulations moved the split across 0.25 to 0.81 within a 0.07 band of perplexity. Granularity buys reallocation; it does not buy it for nothing.

A natural repair would be a single scalar folding k in, the fraction of *active* feed-forward width the shared branch represents, $\phi = I_{\text{shared}}/(I_{\text{shared}} + kI)$. Two further runs test it directly, and refute it. Both hold $k \cdot I = 2048$ and $I_{\text{shared}} = 1024$, the same active widths and the same $\phi = 0.33$ as the reference configuration ($s_{\min} = 0.47$), and change only how the routed width is delivered: as four experts of 512, or as one expert of 2048. If active width were sufficient, both should land near 0.47. Neither does. Four-of-512 reaches 0.29 (final perplexity 3.06 at 96M parameters), well below the reference, so granularity acts on its own even among $k \geq 2$ configurations: the same active width delivered through more, narrower experts engages the routed branch more. At matched active width and a fixed expert pool, this is a controlled test of DeepSeekMoE's fine-grained segmentation argument [1], and it comes out in that argument's favour, on engagement here and on specialization in Section VII. And one-of-2048 stays at 0.95, so top-1's collapse is not a width effect either; doubling the per-token routed width changes essentially nothing when the router cannot mix experts.

The top-1 failure is not mysterious. Single-expert routing removes the router's ability to blend experts, and the routed branch never becomes a useful object at all: at $I = 1024$, ablating it entirely costs 0.29 perplexity, and its domain NMI not only stays near zero but *declines* over training, from 0.0053 to 0.0003. Top-1 is a qualitatively different regime rather than an extreme point of the same one, consistent with the difficulty reported for single-expert routing in the literature [6]. What the data support is therefore a joint statement, not a scalar law: at fixed granularity the width proportion orders every configuration we measured; at fixed active width, finer granularity lowers the shared share further; and single-expert routing never engages the routed branch at any width we tested.

The results of this section compose. The gradient race decides which branch arrives first; the branch proportion and the routing granularity decide where the allocation settles once it is over. The width manipulations are reparametrizations rather than levers on quality, the minimum shared share spanning 0.25 to 0.81 across the seven 127M–234M configurations while final perplexity spans 2.95 to 3.02; granularity is the exception, and the one lever that must be paid for.

VII. Does Reallocation Buy Specialization?

The motivation for wanting routed experts engaged is usually that they are the object of study, so it matters whether experts forced to carry more computation also organize their work more sharply. We measure gate-weighted normalized mutual information between expert assignment and the domain label of each token, comparing at matched training steps and, where possible, at identical routed architecture. Fig. 6 collects the result, and it is not the simple one.

The width lever delivers. Holding $E = 8$ and $I = 1024$ so that the routed branch is identical across runs and varying only I_{shared}, domain NMI at 983M tokens rises from 0.0103

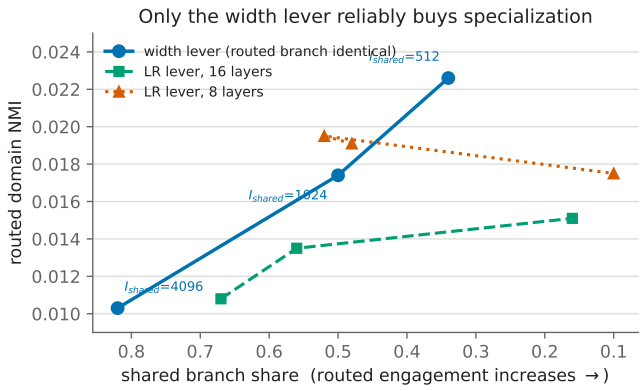


Fig. 6. Routed domain specialization against branch split, with routed engagement increasing to the right; the width series is read at 983M tokens, the learning-rate series at 491M. The width lever, whose three runs share a byte-identical routed branch, more than doubles specialization. The learning-rate lever helps at 16 layers, where the baseline is still shared-dominated, and does not at 8 layers, where balance has already been reached.

to 0.0174 to 0.0226 as the shared share falls from 0.82 to 0.50 to 0.34. That is a factor of 2.2 in specialization obtained without touching the routed experts at all, and without cost in perplexity. Because NMI is measured at 983M tokens, the splits quoted here are the values at that point rather than the trajectory minima of Table III, which is why the same three runs appear with slightly different splits in the two places. The granularity runs of Section VI-A point the same way, their routed NMI rising from 0.0003 at top-1 to 0.0235 at top-4, and to 0.0301, the highest routed specialization measured anywhere in this study, when the same active width is delivered as four experts of width 512.

The learning-rate lever does not, or not reliably. At 16 layers, measured at 491M tokens where the baseline remains shared-dominated, slowing the shared expert raises NMI monotonically, 0.0108 to 0.0135 to 0.0151, all three points from the same 15 000-step sweep. At 8 layers, where balance has already been reached by then, the response is non-monotone, 0.0191, 0.0195 and 0.0175 for $m = 1.0$, 0.3 and 0.1, ending below the baseline; this series reads the long-schedule reference against short-schedule sweep runs, so we take from it only the absence of a reliable gain. The dose-response series of Section V is similarly equivocal: NMI increases with handicap duration among the ramped runs but the unhandicapped baseline falls inside their range rather than below it.

The pattern these disagreements share is that specialization tracks how much the routed branch has genuinely been trained, which rises as it takes over computation but flattens near balance and can fall once the shared branch is starved past it. Engagement and specialization are therefore related but distinct axes, and the distinction is practical: an experimenter who wants routed experts that are better objects of interpretability should change the branch widths rather than the learning rate, even though both move the split by comparable amounts. We note that domain labelling is one specialization axis among several, that our 16-layer learning-rate runs are single-seed, that we deliberately exclude comparisons across expert count here because NMI grows mechanically with the number of experts, and that every value is measured at balancing coefficient 10^{-2}, which suppresses domain NMI relative to unbalanced training, so the magnitudes are conditional on that setting even though each contrast is internal to it.

VIII. Discussion

The picture that emerges is that a small MoE's division of feed-forward labour is only weakly determined by the task. Two quantities that a practitioner sets freely, the relative learning rate of the two branches and their relative width, move the split across almost its entire attainable range while leaving held-out perplexity within 0.1, and a third, the granularity of the routing, moves it further at a cost the first two do not carry. The shared expert is not doing something the routed experts could not do; it is doing what it got to first, and it keeps as much of the work as its width allows.

Three consequences follow. The first is diagnostic. Expert redundancy cannot be assessed from a single early checkpoint, because the measurement at 98M tokens and the measurement at 983M tokens disagree, and deeper models remain in the misleading regime longer. Branch importance should be reported as a trajectory. The second is a matter of experimental design. The earliest version of this study swept expert count at fixed per-expert width and read the outcome as a dependence on E; that sweep let routed capacity and parameter count ride along with the variable under study, and its conclusion did not survive the matched comparisons of Section VI. Sweeps over architecture should fix what they are not testing. The third is prescriptive, and Section VII makes it sharper than we expected. A modest learning-rate handicap on the shared expert engages the routed branch much sooner at no measured cost, and need only be applied over the early part of training; but engagement bought that way does not reliably come with sharper specialization. If the routed experts are wanted merely *used*, either lever serves. If they are wanted *legible*, the width ratio is the one that works. The mechanism also yields a prediction for routing families we did not test: expert-choice routing [21] equalizes token counts across routed experts but leaves the shared expert's every-token exposure intact, so the race, and its learning-rate remedy, should survive it.

Several limits bound these claims, and they concern generality rather than internal validity. All models here fall between 58M and 251M parameters and see at most 983M tokens, on a single data mixture and a single hardware configuration; whether the crossover and the two controls persist at production scale is untested and is the obvious next question. The width-ratio result establishes an ordering over an eightfold range at one depth, not a functional form; the granularity runs show that no single scalar we tried subsumes width and routing at once, and we deliberately do not fit one. The dose-response over handicap duration is replicated at a single endpoint, and its remaining points rest on one seed

each; the effect sizes there exceed the measured seed noise by a comfortable margin, but the shape of the flattening deserves a denser sweep. Finally, our evidence is perplexity, ablation impact and routing statistics. At this scale standard reasoning and question-answering benchmarks sit at or near chance, so we cannot say whether a reallocated split changes downstream behaviour; that is the measurement a practitioner would ultimately want, and answering it at all requires a larger model rather than a further experiment on this one.

IX. Conclusion

Routed experts in small Mixture-of-Experts models are not redundant. They are out-competed early in training by a shared expert that receives gradient from every token, and the outcome of that competition is settled by choices the designer makes rather than by the demands of the data. Slowing the shared expert reverses the outcome durably, the reallocation persisting to the end of our budget even when the handicap covers only the first tenth of training. Where the split comes to rest is then set jointly by the relative width of the two branches, which spans shared shares of 0.25 to 0.81 at essentially unchanged perplexity, and by the granularity of the routing, the strongest lever and the only one that is not free; single-expert routing never engages the routed branch at any width we tested. Where a small Mixture-of-Experts model puts its feed-forward computation is, to a good approximation, something the designer chooses rather than something the model discovers.

Reproducibility

The model, the training code and the analysis toolkit, including whole-branch ablation, routing capture and the trajectory drivers, will be released publicly upon acceptance, together with the run configurations and the per-checkpoint measurements behind every table and figure here. Every experiment in this paper runs on a single GPU. Two implementation defects found and fixed during the study, a missing causal mask in the absorbed attention inference path and a broken CUDA reduction in the load-balancing loss, are recorded in the version history that will accompany that release; every measurement reported here postdates both fixes.

References

[1] D. Dai, C. Deng, C. Zhao *et al.*, "DeepSeekMoE: Towards ultimate expert specialization in mixture-of-experts language models," *arXiv preprint arXiv:2401.06066*, 2024.

[2] N. Muennighoff, L. Soldaini, D. Groeneveld *et al.*, "Olmoe: Open mixture-of-experts language models," in *International Conference on Learning Representations (ICLR)*, 2025.

[3] X. Lu, Q. Liu, Y. Xu, A. Zhou, S. Huang, B. Zhang, J. Yan, and H. Li, "Not all experts are equal: Efficient expert pruning and skipping for mixture-of-experts large language models," in *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.

[4] N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, and J. Dean, "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer," in *International Conference on Learning Representations (ICLR)*, 2017.

[5] D. Lepikhin, H. Lee, Y. Xu, D. Chen, O. Firat, Y. Huang, M. Krikun, N. Shazeer, and Z. Chen, "GShard: Scaling giant models with conditional computation and automatic sharding," in *International Conference on Learning Representations (ICLR)*, 2021.

[6] W. Fedus, B. Zoph, and N. Shazeer, "Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity," *Journal of Machine Learning Research (JMLR)*, vol. 23, no. 120, pp. 1–39, 2022.

[7] B. Zoph, I. Bello, S. Kumar, N. Du, Y. Huang, J. Dean, N. Shazeer, and W. Fedus, "ST-MoE: Designing stable and transferable sparse expert models," *arXiv preprint arXiv:2202.08906*, 2022.

[8] A. Q. Jiang, A. Sablayrolles, A. Roux *et al.*, "Mixtral of experts," *arXiv preprint arXiv:2401.04088*, 2024.

[9] DeepSeek-AI, "DeepSeek-V2: A strong, economical, and efficient mixture-of-experts language model," *arXiv preprint arXiv:2405.04434*, 2024.

[10] ——, "DeepSeek-V3 technical report," *arXiv preprint arXiv:2412.19437*, 2024.

[11] A. Yang, B. Yang, B. Hui *et al.*, "Qwen2 technical report," *arXiv preprint arXiv:2407.10671*, 2024.

[12] A. Achille, M. Rovere, and S. Soatto, "Critical learning periods in deep networks," in *International Conference on Learning Representations (ICLR)*, 2019.

[13] J. Frankle, D. J. Schwab, and A. S. Morcos, "The early phase of neural network training," in *International Conference on Learning Representations (ICLR)*, 2020.

[14] D. Dai, L. Dong, S. Ma, B. Zheng, Z. Sui, B. Chang, and F. Wei, "Stablemoe: Stable routing strategy for mixture of experts," in *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022.

[15] Z. Chen, Y. Deng, Y. Wu, Q. Gu, and Y. Li, "Towards understanding the mixture-of-experts layer in deep learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[16] J. Su, Y. Lu, S. Pan, A. Murtadha, B. Wen, and Y. Liu, "RoFormer: Enhanced transformer with rotary position embedding," *arXiv preprint arXiv:2104.09864*, 2021.

[17] B. Zhang and R. Sennrich, "Root mean square layer normalization," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[18] G. Penedo, H. Kydlíček, A. Lozhkov *et al.*, "The FineWeb datasets: Decanting the web for the finest text data at scale," *arXiv preprint arXiv:2406.17557*, 2024.

[19] L. Ben Allal, A. Lozhkov, G. Penedo *et al.*, "Cosmopedia: How to create large-scale synthetic data for pre-training," Hugging Face Blog, https://huggingface.co/blog/cosmopedia, 2024.

[20] K. Paster, M. D. Santos, Z. Azerbayev, and J. Ba, "OpenWebMath: An open dataset of high-quality mathematical web text," *arXiv preprint arXiv:2310.06786*, 2023.

[21] Y. Zhou, T. Lei, H. Liu, N. Du, Y. Huang, V. Zhao, A. M. Dai, Z. Chen, Q. V. Le, and J. Laudon, "Mixture-of-experts with expert choice routing," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.